

What Does a Sim-to-Real Correlation Support?

A Bounded 26-Paper Reporting Audit and Decision-Level Reanalysis

Tri Lam - 6 August 2026

Abstract

Sim-to-real correlation is often used to support robot-policy evaluation, but the practical question is whether acting on an evaluator selects an acceptable real policy. A coefficient describes association among its entering points; it does not specify the action, loss, or transport to new policies and tasks. We audit 26 papers and find finite displayed panels in all 26; none defines a target population or probability-sampling mechanism.

In this non-systematic set, top-1 decisions agree in 17 of 19 complete matrices. Real2Sim’s checkpoint-selection aggregate has $r = .878-.947$ but selects a different policy, with 12.5 percentage points of displayed real regret. Equal task weighting for Cosmos-Surg gives $r = .883$, Spearman $\rho = .371$, a different winner, and 10.0 points of regret. OSCAR’s plotted values give $r = .855$ with a 1.1-point argmax difference, but its axes come from different evaluation pools and its uncertainty cannot be reconstructed. Rank-aware companion metrics add useful evidence but do not specify the downstream action, acceptable loss, or real-test budget.

High correlation can coexist with task disagreement, reversed checkpoint selection, and metric-implementation discrepancies. Correlation alone does not determine winner agreement or regret.

1. Introduction

Real-robot evaluation is expensive, so simulation, learned world models, and video evaluators are increasingly used to rank policies before deployment. The consequential act is usually not reporting a coefficient; it is retaining, testing, or deploying a policy because of that coefficient. Correlation between simulated and real performance is relevant to that use, but three different claims are often left entangled:

1. simulated and real scores are associated for the displayed points;
2. that association extends to new policies, tasks, sites, or sessions; and
3. selecting by simulation produces an acceptable real-world decision.

The first claim is descriptive. The second requires a target population and a design or model linking the observed panel to that population. The third requires a decision rule and loss. A high coefficient can coexist with a ranking reversal near the top because Pearson correlation summarizes average linear association rather than winner identity or decision regret.

Prior work has examined sim-to-real predictivity, repeated-observation correlation, and reliable reinforcement-learning evaluation [1-5]. Recent robotics systems have made simulation-based evaluation more capable and reproducible [6-10]. Policy selection under uncertainty, top-k regret, and downstream regret are also established ideas [33-35]. We apply them to ask:

What does a reported sim-to-real coefficient support for a specified target, selection rule, and loss?

We contribute:

- a framework separating association, finite-panel action, population transport, and decision acceptability;
- an audit of 26 papers’ correlation claims and a 19-case decision ledger;
- task-composition and measurement sensitivities;
- a reconstruction of six Real2Sim MMRV values; and
- a reporting checklist for future evaluations.

2. Association, population, and decision

Let Y_i^R and Y_i^S be real and simulated scores for displayed unit i in panel $\mathcal{P}$. The Pearson coefficient

$$r_{\mathcal{P}} = \frac{\sum_{i \in \mathcal{P}} (Y_i^R - \bar{Y}^R)(Y_i^S - \bar{Y}^S)}{\sqrt{\sum_{i \in \mathcal{P}} (Y_i^R - \bar{Y}^R)^2} \sqrt{\sum_{i \in \mathcal{P}} (Y_i^S - \bar{Y}^S)^2}}$$

is a property of that displayed panel. The index i may represent policies, checkpoints, tasks, conditions, or a product of axes. A population correlation requires a target distribution over the relevant units and a defensible link from the observed panel to that distribution.

A policy-selection decision requires different quantities. For candidate set $\mathcal{A}$, define the displayed real and simulated winner sets as

$$R = \arg \max_{a \in \mathcal{A}} Y_a^R, \quad S = \arg \max_{a \in \mathcal{A}} Y_a^S.$$

Selection is possibly correct when $R \cap S \neq \emptyset$, and robustly correct when every simulated maximizer is a real maximizer, $S \subseteq R$. Displayed real regret for selected policy a is

$$\max_{b \in \mathcal{A}} Y_b^R - Y_a^R.$$

These definitions preserve ties and expose the decision consequence. They do not convert a convenience panel into a policy population.

WorldGym illustrates why the axes cannot be collapsed into one effective sample size. Its Figure 4a contains three policies repeated across 17 tasks, giving 51 policy-task cells and reporting $r = .78, p < .001$ [10]. The policy axis has three blocks and the task axis has 17; each supports a different question and exchangeability argument.

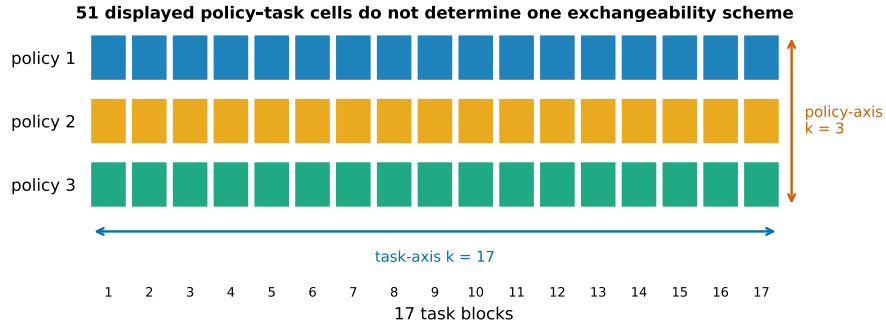


Figure 1: The same 51 cells support distinct policy- and task-axis questions.

The evidence required depends on the claim being made:

claim	minimum object	what remains unresolved
Displayed-panel association	Paired real and simulated values, metric definition, and unit identity	Which policy an evaluator selects and the real loss of that choice
Finite-panel decision	Candidate set, aggregation and selection rule, ties, real winner set, and loss	Performance for new policies, tasks, sites, or sessions
Population transport	Target population plus a sampling, assignment, or predictive model linking it to the panel	Whether the resulting decision is acceptable for the intended use
Decision acceptability	Named action, loss, tolerance, uncertainty, and downstream real-test budget	Broader uses not covered by the declared target and tolerance

Passing an earlier row does not pass a later one. Conversely, a useful finite-panel result need not make a population claim if its action and boundary are explicit.

3. Methods

3.1 Corpus and coding

We included 26 papers that evaluated robot policies, checkpoints, or variants with aligned simulated and real outcomes and printed a named association coefficient. The corpus is non-systematic. We coded each claim’s units, target, transport basis, checkpoint selection, and coefficient uncertainty.

3.2 Numeric data

We extracted numeric results for 19 papers from source data, tables, or figures and checked figure-derived values against reported statistics where possible. Matching a reported coefficient checks extraction, not the underlying experiment. We label figure-derived values and do not infer missing trial counts.

3.3 Decision atlas

For 19 complete matrices, we preserved source-defined rules where available; otherwise we weighted displayed tasks equally, retained ties, and computed displayed real regret. The resulting top-1 rule agrees in 17 cases and disagrees in two.

3.4 Task composition and real-trial remeasurement

We test all displayed-task subsets and leave-one-task-out aggregates; these do not represent future-task probabilities. Where real trial counts are known, independent Beta(1,1) models estimate winner uncertainty. Missing simulator counts and dependence are handled by sensitivity analysis.

4. Reporting audit

result	count	interpretation boundary
Included papers	26	non-systematic corpus
Complete numeric matrices	19/26	source- and figure-derived
Finite displayed panel	26/26	all audited papers
Defined target population or sampling mechanism	0/26	none reported
Coefficient uncertainty	5 p; 1 CI; 20 neither	p-values test a null rather than quantify magnitude uncertainty
Checkpoint or model state selection (yes/no/N/A)	8 / 17 / 1	disclosed in eight papers
Fewer than ten policy or checkpoint blocks	23–25/26	range reflects ambiguous reporting

No paper defines a target population or probability-sampling mechanism, although some make model-based transport arguments. Several papers report held-out prediction, but none defines a future-policy or future-task distribution with calibrated population-level uncertainty. Most reports also do not define the test or exchangeability unit needed for a common inferential interpretation.

5. Correlation and displayed decisions

Figure 2 compares illustrative agreement and disagreement cases at nearby, often high correlations. It is not a prevalence sample; the complete 19-case ledger appears in the supplement.

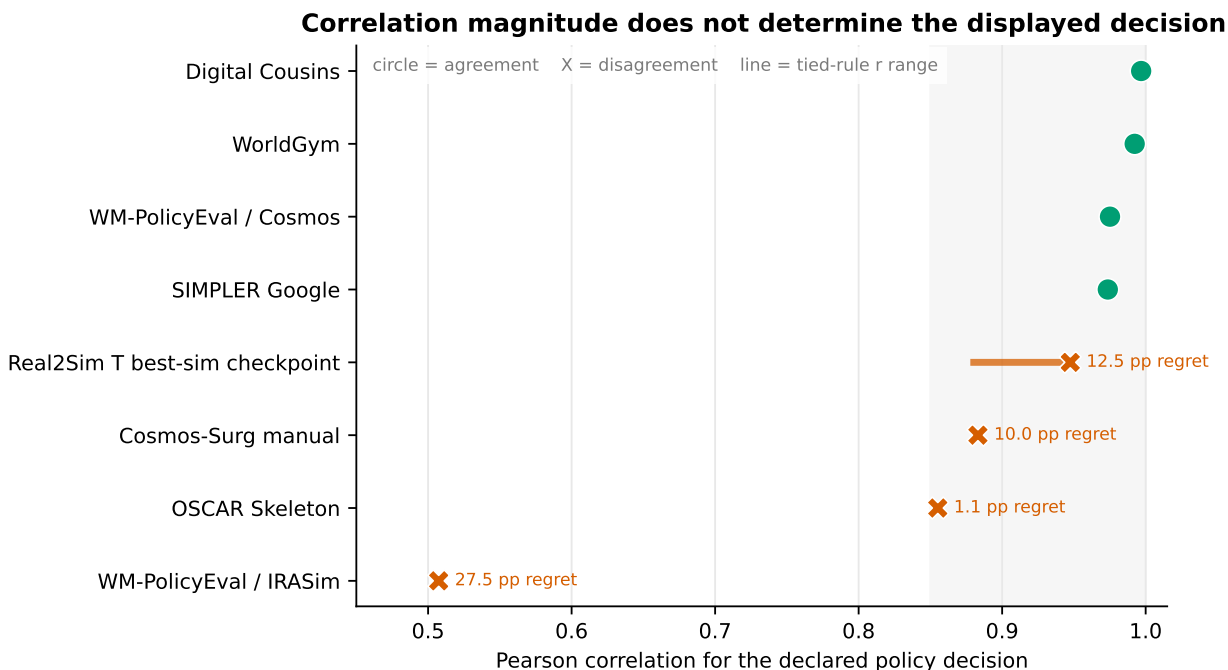


Figure 2: Illustrative cases: similar correlations can yield agreement or disagreement. Labels show displayed real regret. Cases use different units and aggregation rules and are not directly comparable.

case	r	sim best	real best	regret	LOTO
WorldGym	.992	OpenVLA	OpenVLA	0	17/17
Digital Cousins	.997	π_0	π_0	0	4/4
SIMPLER Google	.974	RT-1-converged	RT-1-converged	0	5/5
Real2Sim T (best-sim)	.878-.947	π_0	DP	12.5 pp	–
OSCAR Skeleton	.855	PG-FAST+	pi0-FAST	1.1 pp	–
Cosmos-Surg manual	.883	GR00T N1.5 50k	GR00T N1 20k	10.0 pp	0/4
WM-PolicyEval Cosmos	.975	OpenVLA	OpenVLA	0	4/4
WM-PolicyEval IRASim	.507	Octo-Base	OpenVLA	27.5 pp	1/4

LOTO counts robustly correct leave-one-task-out aggregates. WorldGym reports a pooled-cell $r = .78$; equal policy averaging gives $r = .992$. For Cosmos-Surg, equal task weighting gives $r = .883$; averaging by training lineage gives $r = .905$, $\rho = .500$, and 3.7 points of displayed regret. Real2Sim’s tie changes r but not the selected winner or regret. Rank-aware metrics reveal additional disagreement but do not define the action, loss, tie rule, or evaluation budget.

Aggregate agreement can conceal task disagreement: SIMPLER Google has $r = .974$ and an equal-task winner match, but only 3/5 task winners agree. The evaluation rule can also reverse the result: Real2Sim rope moves from $r = .901$ over all checkpoints to $r = -.683$ after best-simulated checkpoint selection. Figure 2 shows that nearby coefficients can therefore support different actions.

5.1 Agreement cases

WorldGym and Digital Cousins retain the correct winner under every single-task deletion and nearly or fully all task subsets. SIMPLER Google also retains its aggregate winner after every deletion, although only 3/5 task winners agree [6,13]. These examples show stable agreement on their displayed tasks.

5.2 Disagreement cases

Real2Sim motivates simulation for checkpoint selection [7]. Selecting each policy’s best displayed simulated checkpoint in the T block chooses π_0 instead of the real winner DP, with 12.5 points of displayed regret. An SVLA checkpoint tie changes r from .878 to .947 but not the decision. Across three tasks and three tie-complete rules, seven of nine disagree, one agrees, and one depends on tie handling.

OSCAR’s plotted values give $r = .855$ [29]. PG-FAST+ wins on the world-model axis and pi0-FAST on the real axis, a 1.1-point near-tie. The axes use different evaluation pools (65 versus 63 sessions), preventing uncertainty reconstruction. Its printed Spearman $\rho = .750$ and MMRV = .571 are not jointly reproducible from the displayed bars under the published metric definitions; the required per-policy rank inputs were not released.

Cosmos-Surg’s equal-task aggregate selects GR00T N1.5 50k in simulation and GR00T N1 20k in reality, with 10.0 points of regret [18]. Every single-task deletion retains the mismatch. Real outcomes use ten trials per policy-task, without released dependence information.

WM-PolicyEval changes the decision when only the evaluator changes [31]. Cosmos selects the real winner OpenVLA at $r = .975$; IRASim selects Octo-Base at $r = .507$, producing 27.5 points of regret. The plotted points reproduce the appendix values despite an inconsistent IRASim legend, and label-order ambiguity does not change either decision. Simulator counts are unstated.

In *A Practical Recipe*, simulated and real top-1 disagree in 3/11 printed rank panels; absolute regret cannot be recovered [9].

6. Robustness to measurement and task composition

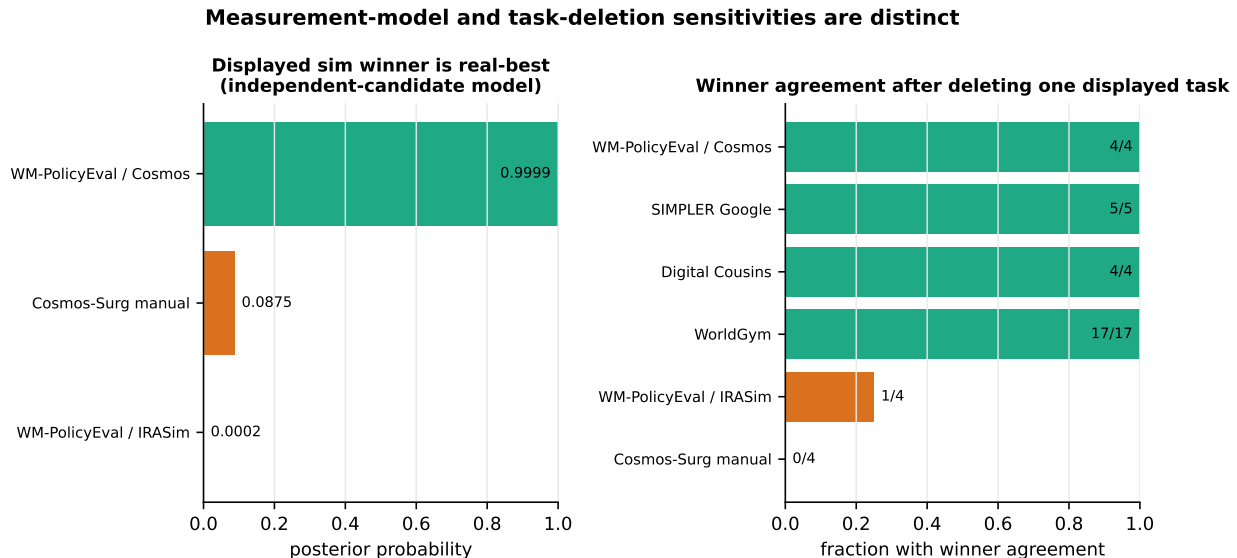


Figure 3: Model and task-composition sensitivities, not calibrated evaluator reliability.

Holding the simulator winner fixed, the Beta-binomial model gives Cosmos-Surg .087 winner concordance and 11.6 points of expected regret. A two-sided sensitivity gives .207–.220 concordance and 8.36–9.65 points of regret; dependence is not identified.

Because WM-PolicyEval omits simulator rollout counts, sensitivity ranges are .333–.9999 for Cosmos and .0001–.432 for IRASim. Policy-specific evidence can reverse IRASim toward OpenVLA; the unknown counts and dependence prevent a reliability estimate.

Both WM evaluators underpredict real success. Positive-slope affine recalibration lowers error without changing rankings. An in-sample isotonic fit changes IRASim’s winner to OpenVLA but uses the same outcomes for fitting and evaluation, so it is only a shape sensitivity.

Task-subset results answer a different robustness question. WorldGym and Digital Cousins remain correct under every leave-one-task-out deletion. Cosmos-Surg manual remains wrong after every deletion. WM-PolicyEval/IRASim is composition-sensitive, becoming robustly correct in one of four leave-one-task-out aggregates. A second deletion is only possibly correct: the simulated winner set ties a real winner with a non-winner. Thus full-panel top-1 alone does not distinguish stable agreement from a composition-dependent result, and tie handling can matter.

7. Metric semantics and numerical provenance

Real2Sim’s Figure 3 correlations reproduce within .0004. Treating one coincident rope marker as one or two points changes r but not any top-1 conclusion.

One tested convention reproduces all six Real2Sim MMRV values: ordering disagreement weighted by the simulated-side gap and divided by N [6].

8. Data and code availability

Data and code are provided with this article. Project materials and checklist: <https://github.com/trilamsr/research/tree/main/sim2real-correlation-audit>.

9. Reusable decision audit and reporting standard

The accompanying checklist covers the decision, data, sensitivities, target, loss tolerance, and real-test budget.

10. Limitations

This non-systematic 26-paper corpus cannot estimate field prevalence and is concentrated in recent manipulation and world-model evaluation. Figure-derived values remain subject to raster resolution, overlapping marks, source revisions, and unlabeled points despite matching reported coefficients.

The cases differ in robots, tasks, candidates, aggregation, source quality, and loss. They do not identify a universal threshold or population selection-error rate. Real-trial estimates are model-conditional; missing simulator counts and dependence limit their interpretation. Task-subset results describe displayed tasks, not future-task probabilities.

Missing target or uncertainty information limits inference; it does not imply evaluator failure.

11. Conclusion

A sim-to-real correlation is a property of a displayed panel. The audited cases show that similar, often high coefficients can accompany different winners and losses under named decision rules. Measurement and task-composition sensitivities then answer distinct questions; neither creates a population claim from a finite panel.

Correlation remains useful evidence of association. It becomes actionable only with a defined target, decision rule, and loss tolerance; estimated outcomes also require decision-relevant uncertainty.

Author and declarations

Tri Lam conceived the audit, curated the records, implemented the analyses, and wrote the manuscript. No conflicts of interest are declared. The paper is a retrospective public-record audit and reports no new human- or animal-subject experiment.

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Supplement to: What Does a Sim-to-Real Correlation Support?

Tri Lam

2026-08-06

S1. Consensus and bounded-sample facts

fact	result	interpretation
Included papers	26	non-systematic claim-based corpus
Papers with recovered numeric results	19	recovery from published figures or tables
Finite displayed panel	26/26	all audited papers
Defined target population or probability-sampling mechanism	0/26	none reported
Coefficient p-value	5/26	null-test output, not interval uncertainty in magnitude
Coefficient interval	1/26	interval uncertainty for coefficient magnitude
Neither p-value nor coefficient interval	20/26	observable reporting fact; not evidence the coefficient is wrong

S2. Unit-count sensitivity

The two columns answer a policy/checkpoint sensitivity question; they do not treat tasks or conditions as automatically exchangeable units.

paper	legacy policy/lineage blocks	permissive checkpoints/variants
real2sim-eval	4	20
RoboWorld	8	8
Digital Cousins	4	4
SIMPLER	4	6
SimFoundry	5	8
WorldGym	3	3
RoboSnap	1	10
REALM	3	3
PolaRiS	4	4
SC3-Eval	1	7

paper	legacy policy/lineage blocks	permissive checkpoints/variants
WorldEval	4	4
A Practical Recipe	5	5
Cosmos-Surg-dVRK	3	6
Gemini/Veo	8	8
DreamDojo	1	6
dWorldEval	1	5
WEAVER	2	2
PlayWorld	18	18
EmbodiedSplat	2	4
MolmoSpaces	3	8
Mem-World	2	2
Colosseum V2	1	1
VISER	1	2
OSCAR	2	7
Hi-WM	2	4
WM-PolicyEval	3	3

Across these two explicit codings, **23–25 of 26** papers have fewer than ten policy/checkpoint blocks.

S3. Axis-specific best-case permutation resolution

These are combinatorial resolutions conditional on valid exchangeability, not exact test results. Tied statistics can make the attained minimum coarser.

paper	printed p	policy k	1/k!	task k	1/k!
RoboWorld	<0.001	8	0.000025	—	—
WorldGym	<0.001	3	0.166667	17	0.000000
REALM	<0.001	3	0.166667	7	0.000198
Cosmos-Surg-dVRK	<0.001	3	0.166667	4	0.041667
Mem-World	<0.001	2	0.500000	5	0.008333

S4. Complete leave-one-unit/point table

The 0.10 column is retained only as a descriptive screen. The continuous movement and deletion unit are the reported quantities.

dataset	points	deletion units	unit	r	max abs change in r	max abs change in rho	>0.10
real2sim-eval	17	4	policy	0.9444	0.006	0.054	no
toy							

dataset	points	deletion units	unit	r	max abs change in r	max abs change in rho	>0.10
real2sim-eval rope	20	4	policy	0.9007	0.029	0.073	no
real2sim-eval T-block	15	4	policy	0.9147	0.048	0.090	no
RoboWorld 10a	8	8	policy	0.9005	0.075	0.042	no
RoboWorld 10b	8	8	policy	0.8762	0.169	0.113	yes
RoboWorld 9a	8	8	policy	0.9888	0.191	0.021	yes
RoboWorld 9b	8	8	policy	0.9438	0.165	0.090	yes
Digital Cousins (unit = policy)	16	4	policy	0.9094	0.013	0.047	no
Digital Cousins (unit = gen. level)	16	4	generalization level	0.9094	0.037	0.063	no
Cosmos-Surg- dVRK automated	24	3	policy run	0.7561	0.073	0.086	no
Cosmos-Surg- dVRK manual	24	3	policy run	0.7180	0.123	0.131	yes
DreamDojo	6	6	point	0.9953	0.010	0.057	no
MolmoSpaces pick	8	8	point	0.9585	0.020	0.000	no
MolmoSpaces open	4	4	point	0.8523	0.617	0.900	yes
MolmoSpaces close	4	4	point	0.9727	0.040	0.000	no
REALM Overall	21	3	policy	0.9187	0.025	0.022	no
REALM Default	21	3	policy	0.8838	0.135	0.057	yes
REALM VB-POSE	21	3	policy	0.9335	0.080	0.187	no
REALM V-VIEW	14	3	policy	0.8902	0.168	0.080	yes

dataset	points	deletion units	unit	r	max abs change in r	max abs change in rho	>0.10
subject paper toy, 200 episodes	17	4	policy	0.8970	0.011	0.095	no
subject paper rope, 200 episodes	20	4	policy	0.9181	0.043	0.088	no
subject paper T-block, 200 episodes	15	4	policy	0.9501	0.037	0.071	no
VISER Octo	4	4	point	0.9988	0.000	0.028	no
VISER	5	5	point	0.8496	0.135	0.300	yes
OpenVLA							
OSCAR	7	7	point	0.8552	0.086	0.132	no
Hi-WM	12	12	point	0.9540	0.014	0.019	no
WM- PolicyEval	12	12	point	0.7193	0.260	0.130	yes
Cosmos							
WM- PolicyEval	12	12	point	0.2772	0.176	0.237	yes
IRASim							

Median maximum absolute change in $r = 0.061$; range 0.000–0.617. The descriptive 0.10 screen selects 10/28 illustrative rows; this is not a corpus-prevalence estimate.

S5. Bounded decision cases

Each row concerns only the displayed policies and the stated aggregation rule. Top-1 agreement means that the simulator-selected winner belongs to the real-data winner set; regret is the displayed real-success gap to the best policy.

case	declared rule	r	top-1 result	real regret	diagnostic
SIMPLER Google	equal-weight mean over 5 displayed tasks	0.974	agreement	0	3/5 individual tasks agree; 5/5 leave-one-task-out aggregates agree
SIMPLER WidowX	equal-weight mean over 4 displayed tasks	0.950	agreement	0	3/4 individual tasks agree; 4/4 leave-one-task-out aggregates agree

case	declared rule	r	top-1 result	real regret	diagnostic
Real2Sim T-block	select each policy's best simulated checkpoint	0.878–0.947	disagreement	12.50 pp	non-winning checkpoint tie changes r but not the selected or real winner
A Practical Recipe	printed top-ranked variant in each rank panel	varies	3/11 panels disagree	—	largest printed Pearson r among disagreements = 0.672

Across Real2Sim’s three tasks and three declared checkpoint-collapse rules, 7/9 cells necessarily disagree, 1/9 robustly agrees, and 1/9 is tie-dependent. These nine cells share data and are not independent repetitions.

The vector extraction contains two coincident rope/DP markers at (0,0). Removing one changes all-checkpoint r from 0.901 to 0.881 and mean-rule r from 0.697 to 0.538, but it does not change any of the three rope rule-level top-1 conclusions.

S6. Complete direct-cell matrices

All eligible direct-cell matrices derived from the retained source inventory under the declared candidate/block rules; outcome-exposed and not a prevalence denominator.

panel	source metric bundle	audit Pearson r	audit Spearman rho	top-1	regret (pp)	LOTO
Cosmos-Surg-dVRK — automated fig1b	Pearson + MMRV	0.941	0.829	correct	0.00	1/4
Cosmos-Surg-dVRK — manual human vs dvrk	Pearson + MMRV	0.883	0.371	wrong	10.00	0/4
Digital Cousins	Pearson	0.997	1.000	correct	0.00	4/4
EmbodiedSplat	Pearson	0.936	0.800	correct	0.00	2/2
—						
mesh-conditions						
Hi-WM	Pearson	0.964	1.000	correct	0.00	3/3
Mem-World	Pearson + p-value	1.000	1.000	correct	0.00	5/5
MolmoSpaces — common-appendix-roster	Pearson + Spearman	0.982	0.800	correct	0.00	3/3

panel	source metric bundle	audit Pearson r	audit Spearman rho	top-1	regret (pp)	LOTO
REALM — Default	Pearson + MMRV	1.000	1.000	correct	0.00	7/7
REALM — Overall	Pearson + MMRV	1.000	1.000	correct	0.00	7/7
REALM — VB-POSE	Pearson + MMRV	1.000	1.000	correct	0.00	3/7
SIMPLER — google robot	Pearson + MMRV	0.974	1.000	correct	0.00	5/5
SIMPLER — widowx	Pearson + MMRV	0.950	1.000	correct	0.00	4/4
WEAVER — CtrlWorld	Pearson + Spearman	1.000	1.000	correct	0.00	5/5
WEAVER — WEAVER	Pearson + Spearman	1.000	1.000	correct	0.00	5/5
WEAVER — WEAVER-FT	Pearson + Spearman	1.000	1.000	correct	0.00	5/5
WM-PolicyEval — Cosmos	Pearson + MMRV	0.975	1.000	correct	0.00	4/4
WM-PolicyEval — IRASim	Pearson + MMRV	0.507	0.500	wrong	27.50	1/4
WorldEval	Pearson	0.996	0.800	correct	0.00	5/5
WorldGym	Pearson + ranking- preservation analysis	0.992	1.000	correct	0.00	17/17

The complete-matrix set contains 17/19 displayed top-1 agreements and 2/19 displayed top-1 disagreements. These counts describe this non-systematic recovered set only.

S7. Illustrative cross-source decision atlas

Heterogeneous finite-panel cases selected for evidentiary roles; not a calibration sample, prevalence denominator, or common population.

case	r	simulated winner	displayed real winner	regret (pp)	LOTO	evidence
WorldGym	0.992	OpenVLA	OpenVLA	0.00	17/17	exact published table
Digital Cousins	0.997	pi_0	pi_0	0.00	4/4	table-validated recovered values

case	r	simulated winner	displayed real winner	regret (pp)	LOTO	evidence
SIMPLER	0.974	rt-1-converged	rt-1-converged	0.00	5/5	official source arrays
Google Real2Sim T	0.878–0.947	pi0	dp	12.50	—	vector-PDF recovery; printed coefficient reproduced
best-sim checkpoint						
OSCAR Skeleton	0.855	PG-FAST+	pi0-FAST	1.10	—	printed one-decimal bar labels; r reproduced to rounding
Cosmos-Surg manual	0.883	GR00T N1.5 50k	GR00T N1 20k	10.00	0/4	vector-PDF recovery; pooled coefficient reproduced
WM-PolicyEval / Cosmos	0.975	OpenVLA	OpenVLA	0.00	4/4	vector-PDF recovery plus exact appendix real values
WM-PolicyEval / IRASim	0.507	Octo-Base	OpenVLA	27.50	1/4	vector-PDF recovery plus exact appendix real values

S8. Finite-real-trial remeasurement

Model-conditional real-trial measurement uncertainty for cases with documented real denominators; simulator values held fixed where simulator rollout counts are unavailable. Candidate/block probabilities are independent in the fitted model; cross-candidate coupling is not identified by published marginal counts.

case	posterior P(sim winner is real-best)	posterior expected regret (pp)	real denominator	simulator denominator
Cosmos-Surg manual	0.087508	11.64	10 trials per policy-task	10 initial states x 3 generated seeds x 2 raters; aggregate allocation/dependence unreleased
WM-PolicyEval / Cosmos	0.999852	0.00	20 trials per policy-task	unstated
WM-PolicyEval / IRASim	0.000168	24.99	20 trials per policy-task	unstated

S8.1 Missing simulator-evidence sensitivity

The effective simulator evidence grid is a sensitivity design, not an estimate of the unreleased rollout denominator.

evaluator	sampled-winner match range	expected real-regret range (pp)	all scenarios below one half?	first listed crossing by prior (.5/1/2)
Cosmos	0.3333–0.9999	0.0–18.6	false	2/2/5
IRASim	0.0001–0.4320	13.2–26.2	true	none/none/none

S8.2 Policy-heterogeneous simulator evidence

The sampled-winner estimand measures latent-rank concordance under one common assumed evidence size, not observed-action reliability. The next rows vary assumed evidence by policy; actual evidence remains unknown.

IRASim scenario	evidence (Octo-Base/Octo-Small/OpenVLA)	latent-winner concordance	MCSE	posterior-mean simulator winner
common_10	10/10/10	0.4302	0.00070	Octo-Base
openvla_10	500/500/10	0.7339	0.00062	OpenVLA
openvla_0	500/500/0	0.9821	0.00019	OpenVLA

S8.3 Probability-level affine calibration

Cell-rate MSE and the empirical individual-outcome Brier score are different estimands. Their exact difference here is the empirical within-cell outcome variance. Positive affine calibration preserves the displayed aggregate winner.

evaluator	mean predicted-real	cell-rate MSE	empirical Brier	intercept	slope	task-held-out recalibrated MSE	winner preserved?
Cosmos	-0.08750	0.02396	0.20417	0.133	0.791	0.02119	yes
IRASim	-0.14583	0.05563	0.23583	0.240	0.402	0.03693	yes

S8.4 Nonlinear calibration and Murphy decomposition

The isotonic map is fitted and evaluated on the same 12 cells. It is an in-sample shape sensitivity, not prospective calibration or selection repair.

evaluator	raw winner	isotonic winner	raw rate MSE	isotonic in-sample MSE	Murphy reliability	resolution	Brier skill vs prevalence
Cosmos	OpenVLA	OpenVLA	0.02396	0.01149	0.01333	0.02082	0.035
IRASim	Octo-Base	OpenVLA	0.05563	0.02131	0.05219	0.02800	-0.114

S9. Exact displayed-task composition sensitivity

Exact sensitivity over every non-empty subset of complete displayed task/condition blocks; not a sampling probability.

case	leave-one-task-out correct	all nonempty subsets correct
WorldGym	17/17	131068/131071
Digital Cousins	4/4	15/15
SIMPLER Google	5/5	24/31
Cosmos-Surg manual	0/4	2/15
WM-PolicyEval / Cosmos	4/4	12/15
WM-PolicyEval / IRASim	1/4	4/15

S10. Checkpoint-selection sensitivity

All tied maxima are enumerated. A range therefore reflects the selection rule itself, not arbitrary file order.

task	checkpoints	all-checkpoint r	best-real r range	tie combinations	best-sim r range	tie combinations
T	15	0.915	0.878 to 0.944	2	0.878 to 0.947	2
rope	20	0.901	-0.471 to -0.103	4	-0.683 to -0.683	1
sloth	17	0.944	0.905 to 0.941	4	0.933 to 0.968	2

S11. MMRV versus correlation stability

MMRV uses SIMPLER’s strict-ordering XOR, real-side gap, and divide-by-N convention. RoboWorld’s real-side leaderboard score is min-max scaled once on the full panel.

dataset	k	MMRV	absolute MMRV range	relative r swing	relative MMRV swing	ratio
Digital Cousins, by policy	4	0.111459	0.020831	1.9%	18.7%	9.8×
Digital Cousins, by generalization level	4	0.111459	0.088890	8.1%	79.8%	9.8×
RoboWorld 10a	8	0.005765	0.003467	15.4%	60.2%	3.9×
RoboWorld 10b	8	0.055067	0.044383	30.0%	80.6%	2.7×

S12. Real2Sim MMRV convention reproduction

The declared grid has 60 named entries but 48 distinct formulas because one predicate pair is pointwise identical by construction. Exactly one distinct formula matches all three Table I values and all three Figure 9 values: less-than-or-equal XOR (equivalently strict-> XOR), simulated-side gap, divide by N.

panel	N	episodes	recovered r	printed r	exact recovered MMRV	printed MMRV
toy packing	17	20	0.9444	0.944	13/170 (0.076471)	0.076
rope routing	20	27	0.9007	0.901	47/270 (0.174074)	0.174
T-block pushing	15	16	0.9147	0.915	13/120 (0.108333)	0.108

The following appendix values use the same recovered convention:

panel	exact recovered MMRV	decimal
Figure 9 toy_packing	21/200	0.105000
Figure 9 rope_routing	307/2000	0.153500
Figure 9 t_block_pushing	209/3000	0.069667

We recover 15 T-block checkpoints from Figure 3; they reproduce Table I’s $r = .915$ and $\text{MMRV} = .108$. Figure 10 reports a separate 12-checkpoint replay subset.

S13. Model-conditional correlation sensitivity

These are equal-tailed posterior intervals under iid bivariate-normal unit summaries and a uniform prior on ρ . They are not assumption-free intervals.

task	unit-level r	n	95% posterior interval
T	0.962	4	[-0.256, 0.978]
rope	0.697	4	[-0.538, 0.901]
sloth	0.984	4	[-0.163, 0.989]